Somewhat Humble Beginnings *Week 1 8/25/2025*
This week I focused on building my digital portfolio and creating a Linkedin account. Right off the bat I was able to connect with a plethora of people due to me having a wide network of connections in the real world already. What excites me about Linkedin is its versatility and ability to aid in the search for a mentor (which I intend to begin in the near future, though I do have a few professors in mind). Additionally, I learned a lot more about ISM as a class and it genuinely made me feel like I made the right decision by interviewing for the course. The freedom of this environment really resonates with me and I am beyond excited for what's to come. I also have made plans to buy the domain pratyushshukla.com so I can have a more professional website. Although I will keep my website for ISM and this one separate as this website will be more so a sleek, modern place for me to showcase my technical skills while still having some fun with the design and layout. Additionally, I worked on a little machine learning side project a bit more this week but I'll reveal more on that in later updates.

Algorithms and Rain *Week 2 9/1/2025*
Most of this week was spent on identifying potential mentors. Dr. Saraswati Sukumar who works at the Johns Hopkins school of medicine is someone who especially interests me. Her lab focuses on using comprehensive gene expression, methylation, sequencing and metabolomics analysis to identify alterations in breast cancer, eventually exploiting these for early detection and therapy. I believe that I will be a valuable addition to her lab due to the applicability that machine learning and deep learning hold in the biomedical space, especially cancer research. Along with that, I also learned more about machine learning algorithms this week, particularly exploring the XGBoost (Extreme gradient boosting) which is a powerful algorithm designed to combine multiple weak models to build a stronger, more robust model. Additionally, this week, I also purchased the domain "pratyushshukla.com" in an effort to capitalize on the availability before someone else took it. I will eventually make this my own personal page to showcase my projects and experience. I had some fun building the initial page with ReactJS and created an interesting node animation (though the actual website does not have anything yet as I have not ported any of my code/configured the DNS settings). Finally, I plan on getting 1-2 positions involving AI & machine learning outside of my ISM work as well in order to strengthen my github page and have more hands-on experience to show when I eventually begin reaching out to mentors. To achieve this I have been asking around and identified some potential opportunities and I am excited to see how they pan out. The recent downpour also gave me time to invest in my own thoughts a little more which I appreciated.

Poetry in Motion *Week 3 9/8/2025*
This week was entirely spent on identifying potential mentors. One person of interest is Dr. Qian Yin at the University of Texas at Austin. She works at the intersection of biomedical engineering and computation and after reading her paper *"Investigating the optimal size of anticancer nanomedicine"* I realized that her innovative approach to real-world problems is incredibly admirable and fascinating. I am planning on sending her an email this coming week and in order to do so I have been asking previous ISM students for the emails they sent in order to get a better idea of what to say. Additionally, I made a robust resume around 2-3 months ago so I am glad that I have that as a resource to show her. On the topic of gaining outside machine learning

experience I am still awaiting a response from some labs and people I know. I believe the way to experience lies in affiliation and using my connections to the best of my ability. This will help me not only right now but also in my future career as one part of academia that is undervalued is the social aspect. After taking a small break while writing this blog entry, I've now begun on the email to Dr. Yin on a separate Google document, however I've also internalized the fact that rejection is normal and I should not expect too much, regardless of how much I admire the professor and their work. I'll be giving more updates on how this pans out in later entries.

Disappointment & Redirection *Week 4 9/15/2025*
This previous week, I sent the email to Dr. Yin however it's been 5 or so days without a response so my hopes of hearing back from her have dimmed a little. Furthermore, I also emailed Dr. Meng Xia due to my interest in her work on LLMs and human computer interaction. Although I have not heard from her either. Though one benefit of this is that I now have my cold email template fully set up and ready to be sent to other professors (although I do intend to make it a little shorter as I think it is a little too long right now). I also personalize each email with what I like about that specific professor's research in order to seem more invested in their work. Overall, it's been an interesting week, I have a list of potential mentors and I am adding to this list every day so it's definitely a fun experience for me reading all the research and viewpoints that these amazing professors have to offer. One day I wish to contribute just as much as them to biomedical engineering and computer science, either in the form of research, teaching, or founding a startup company. In the near future, maybe a professor will finally respond and give me a chance, and there is nothing I want more at the moment.

Moving at Lightspeed *Week 5 9/23/2025*
This week, I sent emails to three professors, Dr. Park at Rice, Dr. Yuan at UNT, and Dr. Kim at Texas A&M. I am waiting on a response but I also plan on sending out a lot more emails this week. I finalized a cold email template that keeps things short but also explains a little about me as a person. I did this in an attempt to get better outcomes via email. Furthermore, I am also excited for the upcoming ISM symposium on Wednesday of this week. I cannot wait to meet new people and explore a wide array of opportunities. In particular, I am excited to talk to those with experience in either biomedical engineering or computer science (though preferably it would be fantastic if I could find someone that has a mix of both). Mentor searching is a long and strenuous process and I'm beginning to understand that I need to increase the volume of emails I send out because currently I have a 100% rate of not getting a response back. I will also diversify the departments of professors who I email. For instance, instead of just emailing professors working in biomedical engineering I will also email computer science professors with a focus in biotech and professors in other disciplines of engineering that are working at the intersection of ai and healthcare.

Socializing *Week 6 9/29/2025*
This week we had our ISM symposium and I was able to network with a plethora of professionals and others in fields that somewhat pertained to my topic. My most memorable conversations were with Mr. Withrow, a Google Software engineer who had spent the beginning of his career working on various technical aspects of Google products and is now in more of a

management role. While this does not directly tie into biomedical computation, it certainly gave me a glimpse into what could come if I choose to focus on machine learning in college. He also explained that the social dynamic is something that most people miss despite its incredible importance in most workspaces. In the current day, artificial intelligence has claimed so many aspects of the everyday job that one of the only things left for humans to excel at is our own humanity and social skills when interacting with others. The other memorable conversation I had was with Mr. Srikanth Gelly, a CEO in the hardware and software industry. This conversation taught me a lot more about early-stage ventures and how the technology sector can change lives through the amount of freedom it provides. Once again this did not relate directly to my topic but it certainly opened my eyes to the limitless possibilities that lay in front of me. Another thing I would like to discuss is my interviews, both went fantastic and I scored the maximum points in each row. I surprised myself because the only preparation I had going into the interviews was a deep understanding of my own resume and a history of participating in technical discussions. I believe that these two things combined have pushed me to the point where I am now incredibly comfortable with answering hard questions such as “What is your greatest weakness?” and spinning them into a strength, allowing the interviewer to notice my social dexterity. I also have around 20 emails that I will be sending this week to postdoctoral scholars at schools such as Stanford, Texas A&M, and UNT. The most recent email I sent was to Dr. Saraswati Sukumar who works in the field of oncology at Johns Hopkins Medical School. It would be fantastic if she agrees to be my mentor as I already have a connection to her since my friend also was mentored by her. That essentially summarizes my week and I am beyond excited for what lies ahead.

Telegram. *Week 7 10/6/2025*
This week in ISM I focused on developing my technical skills a little bit more. I have extensive knowledge of C++ however I decided that it wouldn’t hurt to give myself a refresher of the language that I haven’t touched in a month or two. I began by looking over some documentation and reading over certain chapters of *The Competitive Programmer’s Handbook* just to give myself a better idea of what exactly I had forgotten. After this I used a resource known as learncpp.com to review some basic syntax. This left me feeling good about myself and helped me regain some self confidence. On a mentor side of things, I revamped my LinkedIn to include more information about myself such as my skills, had my resume reviewed by some of my friends in college and an ISM 2 student (thank you Prisha), and also messaged Dr. Julie Park on LinkedIn to politely ask if she had seen my email. Sometime this coming week I plan on sitting down and sending emails the other potential mentors I have found all in one go, so I am incredibly excited for that, although my hands might not be. Additionally, through a friend at Reedy I learned that I might be working at Bank of America soon as a software engineering intern as his mom is quite high up the ladder there and got him started in a similar role (to which he later offered to do the same thing for me, of course I graciously accepted). My weeks have been getting busier as of recently, but a recent Lex Fridman podcast featuring Telegram CEO Pavel Durov taught me a lot about this. The first thing he stated is that phones are useless, a controversial take but I’ll try to minimize how much I use my device. He also said that he likes to allocate as much time for sleep as possible. I plan on doing this by working hard enough to where I have no work left by a certain time in the day in order to fix my sleep. Finally, the one

thing that stuck with me the most was his idea of freedom being essential. In the future I want to follow a line of work that gives me the freedom I crave while also making a meaningful change in the world, this is my ultimate goal. That's all for this week, thank you for reading, and thanks Pavel!

Game, set, and match. *Week 8 10/6/2025*
I watched a lot of tennis this week, something about the sport has just always excited me; maybe it's the sound the pros are able to create as the ball blasts off their strings, or maybe it's the miracle moments where the underdog comes through and produces a thriller of a set. Speaking about thrillers, Stanford definitely gave me one this week. I emailed three Stanford postdoctoral scholars and… drumroll please… all of them responded! One told me to reach back in January for a potential research position at his lab and another told me that while he was extremely happy to see my dedication he simply did not have any projects at the moment. However, my biggest victory was the third, Dr. Gauri Desai who works at the FASTR lab (a computational biomechanics lab meant to help support underrepresented athletes). My email to her was simple yet intricate. I detailed a paper of hers that I loved, explained who I am, and stated that I would be excited to have the opportunity to work under her, ending with an invitation to have a chat over Zoom. To be honest, I did not expect much at all, but then she emailed back. She said she'd be more than happy to talk to me and asked me for times at which I am available. My first thought was to fall out of my chair, my second was to send a thoughtful reply back. So here I am, awaiting my interview with a Stanford researcher on the 23rd as I attempt to secure my place at their lab. It still feels unreal but I know that the effort does not stop here, in the next week I plan on doing everything I can to prepare myself for this interview such as writing tailored interview questions, reading her research, and diving deeper into the lab's work so I have more talking points. Besides that I've been hard at work coding this past week for a couple personal projects, nowadays I have so much work it all seems like it's going by so fast, but I truly do find joy in making time for the small things I enjoy, whether that be tennis or training AI models.

Sleepy computer science. *Week 9 10/21/2025*
This week I conducted a lot of research on the topic of neural networks and deep learning. Particularly convolutional neural networks and transformers. The difference between these two lies in the efficiency of the various tasks they complete. Convolutional networks are particularly strong at image processing where localized relationships are required whereas transformers shine in developing contextual relationships and long-distance outputs. For the purpose of my original work, I will likely end up using a CNN due to the high volume of image processing that I will be undertaking and the localized pattern-based recognition I will likely employ. With days getting shorter and workload increasing I feel as if I'm getting less sleep which troubles me but realistically there is nothing I can do about it currently. The only way forward is to begin doing my work at an absurdly high pace and thinking about homework all the time which I do my best to achieve. I wonder how the work is at MIT, probably a lot more exhausting than what I deal with right now but maybe it'll be more fun? I have an interview with Dr. Desai on Thursday which I am doing my best to prepare for along with everything else.

The Precipice of Victory. *Week 10 10/27/2025*
This week I had a very insightful discussion with Dr. Desai and she agreed to do her best to get me to do research on one of her projects. She asked for a letter explaining what ISM is alongside some of my past work. For my past work I will send her my GitHub account which has all my past machine learning project repositories stored and available for others to view. The "letter of recommendation" was given to me today by Mr. Ratliff (thank you very much!) and now I am building an email to send to her giving proof of my machine learning experience. I am incredibly excited to see where this leads me in both my research and ISM journey as I begin to truly immerse myself into the world of biomedical computation. Additionally, this week we received our PSAT/NMSQT scores and I scored a 1500 out of 1520. I got a perfect score on the math section and a 740/760 on the reading section which means I performed essentially how I expected to in my quest to achieve national merit semifinalist. I am incredibly happy with my cutoff but the achievement is overshadowed by the mountain of work looming above me alongside my research aspirations. The days have also begun to get colder and I am excited for the walk home to reach freezing temperatures. Winter is definitely one of my favorite seasons.

Infinity. *Week 11 11/3/2025*
Somehow no matter how much work I complete my list of action items seems to grow bigger by the minute. I have become semi-accustomed to this though, right now is arguably one of the most exciting times of my entire life. I talk to new people almost every day (LinkedIn is surprisingly great for talking to people). Besides that, I am proud to announce that I have finally completed my first project using XGBoost (extreme gradient boosting). I used breast cancer data from UCI in order to develop a machine learning model that can predict if a cancer is benign or malignant and tested it as well. The model has surprisingly high accuracy and I plan on fine tuning it even more in the future and perhaps making it into a viable tool using a library such as Streamlit. As research speeches are also to begin this week I am even more excited to discuss my creation and more about what motivates me. On a side note, I have also contacted Dr. Desai at Stanford with my past work and ISM letter and am now awaiting a response. Given her busy schedule, it might be a little before she gets back to me but that is perfectly fine. This year I'm not only doing things that I did not believe I could do before but I am also learning more about myself which I had never thought to do in the past. Finally, Mariah Carey has officially begun defrosting.

The Ball Keeps Rolling. *Week 12 11/10/2025*
As Thanksgiving break draws closer, there are a few important things that have been going on in my life. From an ISM standpoint, I am still waiting on Stanford to give me clearance to access the data I need in order to conduct research in the FASTR lab, additionally my research speech exceeded my expectations and I would say the only things I need to work on are abstracting more technical aspects of my work for a less familiar audience and making my closing stronger. Furthermore, my partner and I have now begun the wait of results for the Congressional App Challenge which we participated in and developed a community driven application that would benefit the elderly population in our area. Moreover, I also have started preparing for the United States computing olympiad which will hold its first round in December. My goal is to be ranked within the gold category and then qualify for platinum later in the year. Even if I don't reach

these goals however, I will have significantly increased my knowledge of computing and algorithmic thinking, both incredibly critical skills to my future. Something I also forgot to mention in previous reports is that my group and I qualified for the international career development conference (ICDC) by achieving first place within the entire Southern region for a financial simulator this past October. So far, this year has been exciting, challenging, and rewarding. I enjoy everything I have learned and am eager to continue and learn even more with an overarching goal of constantly pushing my boundaries.

Acceleration. *Week 13 11/17/2025*
While waiting on a response from Dr. Desai and Stanford I realized it would not be to my benefit to sit idle and let this time go to waste. Therefore, I have began messaging others such as Dr. Gumaro Redon who is a quantum algorithms scientist at Fujitsu Research working primarily in deep learning. I believe by expanding my facets of computer science I will be able to obtain some amazing opportunities and I want to explore these as much as I can. Additionally I have continued to prepare for the United States Computing Olympiad which starts this December. My goal is to achieve gold by March and aim for platinum in the U.S. open competition which happens around June. In doing so I am not only furthering my programming abilities but also setting myself up for success with technical interviews in the future. Thanksgiving break is also drawing near and I have a lot of big plans, I believe in being rational and pragmatic, so if I do not have a response from Stanford by the end of this coming week I will once more start sending out an absurd number of emails each day. I believe that my resume is strong and augmented well by what I have achieved this year so I am excited for the process of continuing to look should it come to that.

Christmas Part 1 *Week 14 12/1/2025*
This week I focused on emailing more researchers and perfecting my breast cancer prediction model for my original work. The model is currently giving accurate predictions that I am decently happy about, however my goal is to transform it into a web-application using either Streamlit or Flask. Additionally, I have also began to finalize my work for my DECA and BPA projects, even though I have already qualified for ICDC this year through the virtual business challenge I would still like to qualify once more with my partner Suhan so that I can be 3x ICDC qualified including my freshman year. BPA is relatively new to me but my partners and I have been making good progress on our paper. I also was able to meet up with a friend who currently goes to Duke over Thanksgiving break and it was a good opportunity to learn more about the school and its culture and I really enjoyed our conversation. It’s been an exciting year so far and although my questions about my mentor are still in the air I believe I am getting close to a solid research contribution role where I can make a meaningful impact. It’s also time to begin thinking about preparation for the USA computing olympiad!

Christmas Part 2 *Week 15 12/8/2025*
This week was centered around emailing even more researchers, I was able to get in contact with a professor at UT Southwestern’s biomedical engineering department and although she said that she did not have any availability, this now reassures me because I did end up getting a response, and I also asked her if she could tell me the names of any acquaintances of hers that

would be interested in allowing me to serve as a research assistant and I am not waiting on a response. In addition to this, I have emailed professors at UNT and Yale and am currently awaiting a response from them. I have at this point made it my personal mission to find a mentor in as less time as possible, while I did have professors tell me to reach out in January and in the Spring from Stanford and Johns Hopkins respectively, I do not wish to wait that long to begin meaningful work and research. I can handle most of the computational aspects of research and machine learning so while creating original work is not a problem, I do wish to partake in laboratory research for the betterment of the world as a whole. Additionally, I have begun preparing for the physics olympiad alongside USACO. It's starting to get intense, but that just means I have to work harder.

The Return *Week 16 1/7/2026*
This week I worked on my posterboard by planning, purchasing materials, and looking at old posters. Over the past week I also caught up on sleep alongside doing some programming and research, fixing bugs in my original work, and also working on my own projects. Additionally, I prepared for the United States Computing Olympiad which is taking place this Saturday and I also prepared for DECA districts with my partner Suhan. Overall my winter break was pretty busy but restful at the same time (even though I was sick for the first few days). Going forward, I plan to finalize my posterboard layout and begin assembling it, while continuing to refine my code and research. I also want to stay consistent with my competitive programming practice after USACO and continue preparing for upcoming competitions. These next few months will be an incredibly busy time for me but I am incredibly glad that I got my SAT and PSAT done with great scores early this year so I do not have to stress about those any more. Now, I can focus on research, competitions, and ISM and remain grounded as I work towards meeting more people and developing my skills across multiple disciplines.

Work, work, work! *Week 17 1/12/2026*
This week I focused on reaching out to more areas where I might be able to contribute. I do have a mentor however I would still like meaningful research opportunities. One of the most exciting upcoming events is my interview with Dr. Xie, a bioinformatics researcher at UT Southwestern whose work focuses on machine learning applications in biology and medicine. I plan to prepare thoroughly by reviewing her recent research, refining my questions around deep learning in biomedical contexts, and thinking critically about how her work connects to my own interests in computational medicine. This interview is a meaningful opportunity to learn directly from a researcher working at the intersection of AI and healthcare. Alongside this, I made progress on my AP Research project, where I am exploring deep learning approaches for medical applications, something that ties into my ISM work heavily. This week, I aim to refine my literature review, clarify my research gap, and continue iterating on my methodology so that my project remains both technically rigorous and impactful. I will also be balancing competitive and extracurricular commitments. After qualifying for BPA regionals with Khajan and Pratham, I plan to begin targeted preparation for the next round, focusing on strengthening our weak areas and improving efficiency. This week has an incredible amount of work but as always I want to come out stronger than when I began.

Outliers *Week 18 1/19/2025*
This week I focused on strengthening my knowledge of kidney cancer and reviewing computational concepts. Dr. Xie informed me that as I am being onboarded to her lab I would have to complete an ethics training before I begin research. She also told me that there would likely be an orientation for new postdoctoral researchers and lab members (including myself!) sometime in the following week. I am beyond excited to begin my research journey and see how I can make a real difference within the world of biomedical computation and bioinformatics. Aside from research, in my GT American Studies class I began reading a book titled *Outliers: The Story of Success* by Malcolm Gladwell. From what I have read so far, the book has opened my eyes to the reality of greatness. I learned that success is the culmination of various factors and it is not simply the byproduct of raw talent. I aim to apply this concept to my own life by opening as many doors as possible for myself. Talking to people, working alongside others more knowledgeable than I, and appreciating multiple perspectives are some methods I plan to use in order to come closer to finding my own definition of success. Overall through busy weeks I've been able to make progress and find structure and I desire to continue doing so.

Freeze *Week 19 1/26/2026*
This week, I stayed focused on building momentum across both research and competitive programming. Recently, I earned Silver in the USA Computing Olympiad, which was a milestone I'm proud of because it reflects consistent effort and measurable growth. With the January contest being one of my last opportunities to advance, I'm now directing most of my practice toward the specific skills I'll need to reach Gold such as speed, accuracy under pressure, and recognizing common patterns quickly. I plan on also mastering certain algorithms that have been prevalent in previous Silver division contests. Even when problems feel overwhelming at first, I've been trying to approach them methodically. The process has been challenging, but it's also been a reminder that progress comes from repeated exposure to difficulty, not avoiding it. On the research side, I continued learning more about kidney cancer and computational approaches in bioinformatics so I can be ready to contribute meaningfully once my onboarding is fully complete. I was also scheduled to have an interview at Rice to potentially gain additional research experience, but it was postponed due to the freeze. While that delay was disappointing, I'm treating it as an opportunity to prepare even more thoroughly so I can make a stronger impression when it's rescheduled.

The Trek to Greatness *Week 20 2/2/2026*
This week, I continued to push forward with a clear focus on one major goal: reaching USACO Gold. Currently being in Silver has been a validating milestone, but it's also made the gap to Gold feel much more concrete. My preparation has become more intentional, centered around sharpening problem-solving speed, improving implementation accuracy, and developing a stronger instinct for identifying patterns that appear repeatedly in Silver-to-Gold–level problems. I've been spending more time dissecting past contests, not just to solve problems, but to understand why certain approaches work and how to recognize them faster under time pressure. The grind has been intense at times, but I'm learning to stay disciplined even when progress feels incremental, knowing that consistency is what ultimately compounds. In parallel, I'm still in the onboarding phase at UT Southwestern Medical Center. While I'm not fully

contributing yet, I've been using this period to familiarize myself with the broader research environment and deepen my background knowledge, especially in areas related to biomedical computation and data-driven approaches to disease. Although I'm eager to start hands-on work, I recognize that onboarding is an important foundation, and I want to be fully prepared to add value once I'm officially integrated into the lab. Overall, this week reinforced the idea that preparation often happens quietly, before visible results appear, whether that's in competitive programming or research.

Time for Competition *Week 21 2/9/2026*
On the ISM side this week I worked on finalizing my speech and also carrying out some basic computational tasks for my mentor which mostly involved cleaning data. Otherwise, the week marked a shift toward balancing algorithmic rigor with fundamental physics, as I continued preparing for USACO while also dedicating time to F = ma. On the USACO side, my focus remained on closing the Silver-to-Gold gap, but I've started approaching problems with a more structural mindset. Instead of jumping straight into implementation, I've been forcing myself to articulate the core idea of a solution before touching code. This strategy definitely played a role in slowing me down initially, but it's already improving my accuracy and reducing the number of dead-end implementations. I've also been paying closer attention to patterns that recur across Gold-level problems, especially those that reward clean modeling over brute force, and training myself to recognize these under time pressure. In parallel, F = ma preparation has demanded a very different kind of discipline. Unlike USACO, where abstraction and algorithms hold great importance, physics punishes any gaps in conceptual understanding. I've noticed that the strongest problems often hinge on translating a physical situation into the right free-body diagram or constraint equation, and small misinterpretations can cascade into completely wrong solutions. The process has been humbling, but also rewarding, as I can feel my intuition for physical systems sharpening.

Exhaustion and more code. *Week 22 2/16/2026*
This last week, I had perhaps the busiest 4 days this entire school year, and truth be told, I didn't live up to my expectations. Each day I felt overwhelmed with work, not knowing what to do when, losing track of deadlines. However, I think if I can find some structure in the craziness of these next few months I'll be just fine. I have amazing teachers and friends and my mentor is also someone I look up to a lot. Speaking of that, my internship at UT Southwestern officially has begun and I am working on a joint project with the kidney cancer department and the quantitative biomedical research center. Admin gave me a UT Southwestern email address and it made me feel incredibly good about myself. It's funny how the small things matter. I am incredibly fortunate to have found a mentor who is not only willing to help me with my ISM product but also have me intern at their institution. Adding onto this, there is another USA Computing Olympiad competition this week and it's my last chance to advance from silver to gold. I'll try my best, but even if I don't, at least I have silver. After all, these next few months all I really can do is try my best after all!

Research time! *Week 23 2/23/2026*

This week things have only accelerated. I have officially started working with Dr. Xie to develop a custom cBioPortal specifically designed for kidney cancer research. At first I did not fully understand what that meant. Now I am learning how large genomic datasets are cleaned structured and uploaded so researchers and clinicians can explore them more easily. We are tailoring the portal to better visualize kidney tumor subtypes mutation frequencies and clinical data. I am starting to see how computational tools directly support real medical decisions. It feels surreal to move from classroom programming assignments to working with real patient derived sequencing data. Each time I log into my UT Southwestern account and test a new configuration I feel like I am contributing to something meaningful. At the same time I began an internship at Stanford where I am working on large language models. The contrast is intense. In one setting I focus on cancer genomics infrastructure and in the other I explore how language models learn patterns and generalize across tasks. Balancing both roles has been overwhelming but also exciting. It has helped me realize that I do not just want to write code. I want to build systems that create real impact in research and beyond.